

Probability Refresher

Stanford University · MS&E 152 · Decision Analysis · Summer 2026

Everything You Need Before the Probability Lecture

What this guide is. The class lecture on probability will move fast and assume you are comfortable with certain ideas. This guide builds those ideas from scratch — starting from the question "*what even is a probability?*" and ending at the rules you will use every day in MS&E 152. No calculus, no prior probability course required. Work through each section in order; every concept leads into the next.

The one idea that runs through everything: probability is how you express **uncertainty**. It is a number between 0 and 1 that says how likely you think something is, given everything you currently know. When you learn something new, those numbers update. That updating is what MS&E 152 is built on.

Your learning path

- §1. Events — what exactly are we talking about?
- §2. The outcome space — listing all possibilities
- §3. Combining events — OR, AND, NOT
- §4. Probability as a measure — the three rules
- §5. Computing with probabilities — computed from counts, conditioning on events
- §6. Independence — when does one thing tell you about another?

§1. Events — What Exactly Are We Talking About?

Start here: what is an event?

Before we can assign a number to something uncertain, we need to be precise about what that "something" is. In probability, we call it an

event — a **statement** (a **proposition**) that is either true or false at a specific point in time. Events are defined by **distinctions**: a 'crisp' yes/no question about the world. An event has two possible outcomes.

Definition (Drake §1.2): event — a statement, a proposition that creates a distinction, at a point in time. 'Point event' means an elemental outcome - the finest level of distinction.

Three examples, going from simple to the kind you'll use all course:

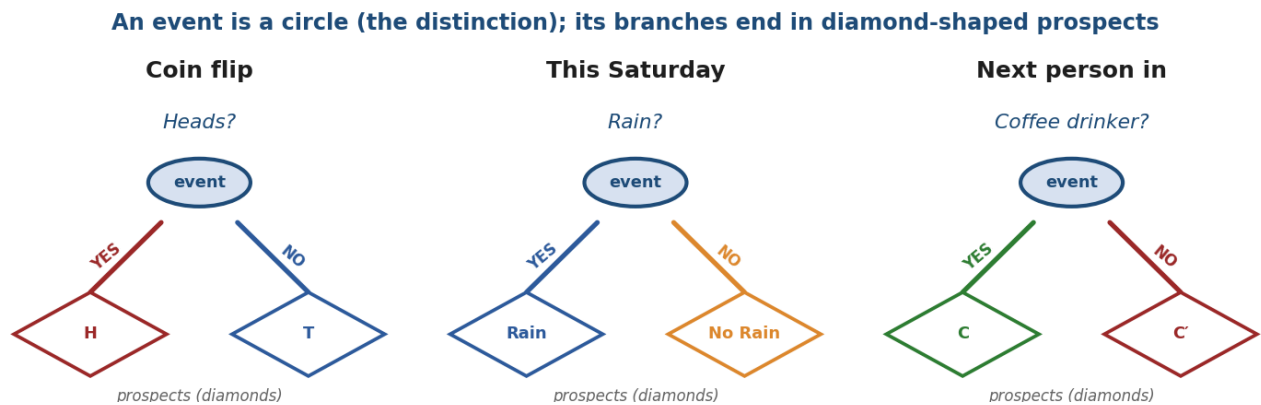


Figure 1. Each event is drawn as a small tree: a circle (the distinction) whose branches end in diamond-shaped prospects. When the moment arrives, exactly one branch is true.

Why be so precise?

In everyday speech, "something might happen" is vague. Probability requires precision:

- **Which** thing? (the distinction)
- **When?** (the point in time)
- *How sure am I?* (the number we'll assign) — this is a **probability measure**.

Intuition: Think of an uncertain event as *something you can bet on*. "It will rain this Saturday" is a clear bet — on Sunday morning you know who won. "The weather will be bad" is not clear enough: bad by whose standard? By when? A well-defined event always has a crisp answer when the moment arrives.

Two symbols to keep apart: $\emptyset$ and ε

The symbol $\emptyset$ is the **empty set** — the impossible event, which can never be true (you'll see it in the axioms in §4). Do not confuse it with ε (epsilon), which stands for the "event" of your background knowledge: the state of information every probability is conditioned on. That is why $P(A)$ always means $P(A \mid \varepsilon)$.

§2. The Outcome Space — Listing All Possibilities

Point events and the outcome space

The most basic events are **point events** (Drake's term) — also called **outcomes** or **elemental outcomes**. These are the most specific things that could happen: you cannot break them down further.

"point event" → elemental outcomes of the outcome space.

outcome space → the complete set of all possible point events. **compound event** → a combination of elemental events (point events).

Two notations for the same idea:

- **Drake / B&H:** The entire "outcome space", equivalent to the "certain event" — denoted Ω (capital omega)

- **Standard:** "sample space" — same symbol Ω

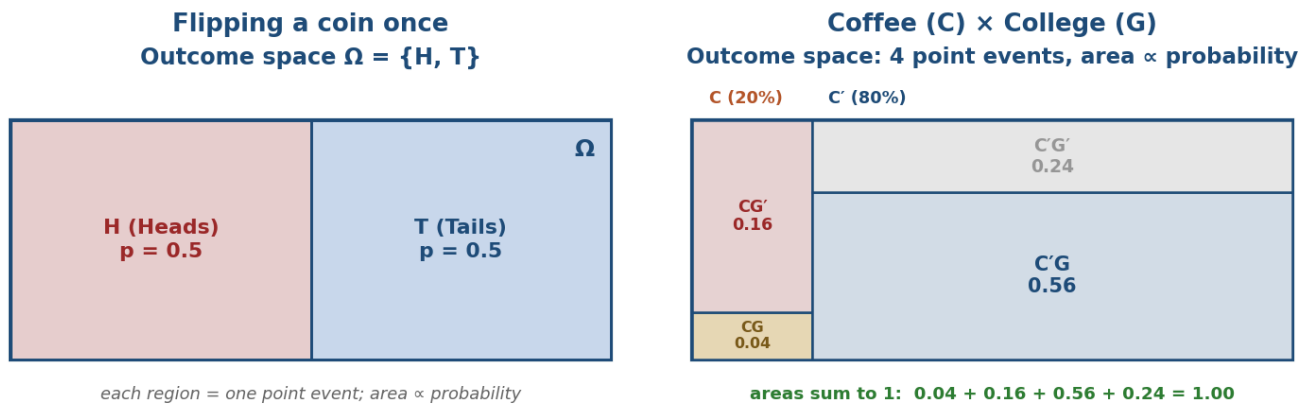


Figure 3. The outcome space as Howard's area diagram: Ω is a rectangle partitioned into point events whose areas are proportional to their probabilities. Left: one coin, $\Omega = \{H, T\}$. Right: Coffee (C) $\times$ College (G) — four point events whose areas sum to 1.

Venn diagrams

Venn diagrams represent the outcome space visually. Each region is a set of point events; its **area is proportional to its probability**.

Venn diagrams are useful for seeing OR, AND, and NOT **Event algebra** expressions geometrically. But for problems with many distinctions, they get crowded. That's why we use probability trees instead.

Probability trees — the main tool of MS&E 152

Probability trees take the place of experiments (Drake §1.2). **Each node is an event** — a distinction at a point in time. **Each leaf is a point event** — a complete, specific outcome.

How to build one:

Step 1: Start from a root node. Ask the first yes/no question (distinction). Draw two or more branches. (A variable may have multiple degrees of distinction, not just binary.)

Step 2: At the end of each branch, ask the next question. Draw two more branches. Repeat.

Step 3: The probability on each branch is a conditional probability (see §5). Multiply all branch probabilities along a path to get the leaf probability.

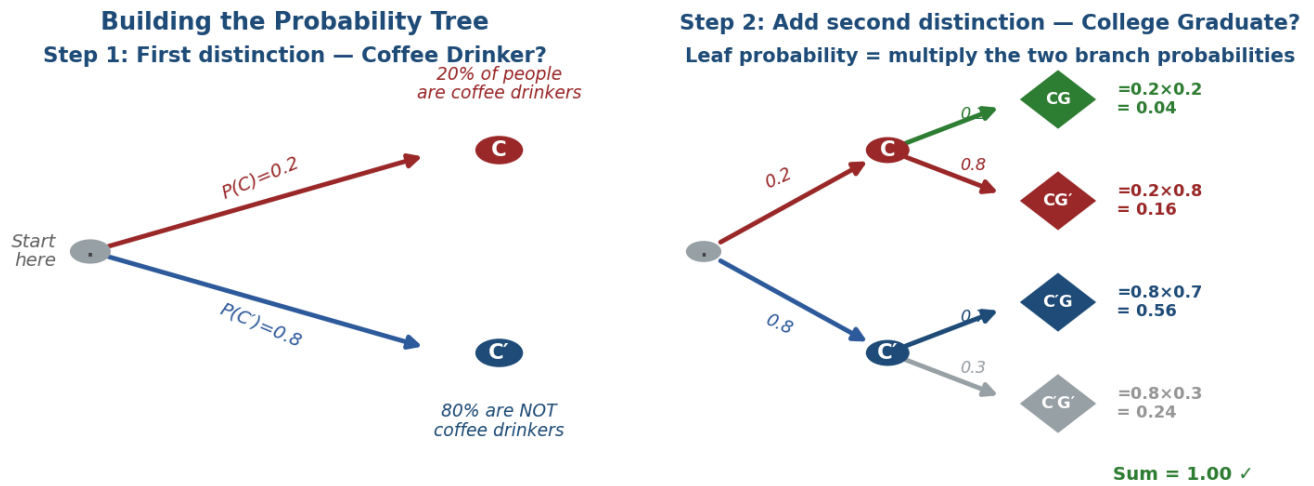


Figure 4. Building the Coffee / College Graduate tree in two steps. Left: first distinction (Coffee Drinker?). Right: second distinction added (College Graduate?). Each leaf probability = product of the two branch probabilities.

Discrete variable: A variable that contains a finite list of possible outcomes. Drake notes:

- A **"binary"** variable has exactly 2 branches $\equiv$ 2 events (e.g. C vs. C^c)
- A variable with 3 outcomes has 3 branches, and so on.

Why trees beat Venn diagrams for this course: Venn diagrams get messy with more than two distinctions. Probability trees scale cleanly to 3, 4, or 10 outcomes. Every problem you solve in MS&E 152 will involve a tree.

§3. Combining Events — OR, AND, NOT

Once you have individual events, you can build more complex ones using three operations. Drake calls this

event algebra — just like algebra lets you combine numbers with +, $\times$, etc., event algebra lets you combine events with OR, AND, NOT.

Event algebra — OR, AND, NOT Set algebra: $\vee$ (OR = union $\cup$) $\wedge$ (AND = intersection $\cap$) X^c or A' (NOT = complement)

The three operations in plain English

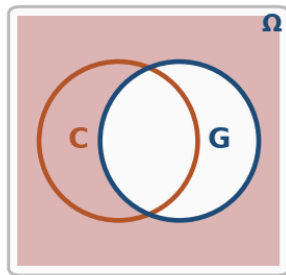
NOT (complement, written A^c): Everything that is *not* A. If A = "it rains," then A^c = "it does not rain." The complement is the flip side.

AND (intersection, written $A \cap B$): Both A and B are true. If A = "coffee drinker" and B = "college graduate," then $A \cap B$ = "coffee drinker who is also a college graduate."

OR (union, written $A \cup B$): At least one (and possibly both) of A or B is true. "Coffee drinker or college graduate (or both)."

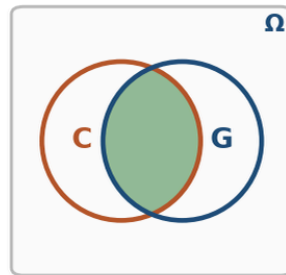
Combining Events with OR, AND, NOT (Venn diagrams on C and G)

NOT college grad (G')



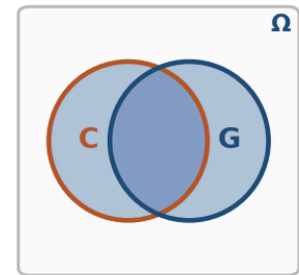
$$G' = \{CG', C'G'\}$$

Coffee AND Grad



$$C \cap G = \{CG\}$$

Coffee OR Grad



$$C \cup G = \{CG, CG', C'G'\}$$

Figure 2. The three event-algebra operations as Venn diagrams on circles C and G inside Ω . The shaded region is the event G' : NOT — everything outside G^c ; AND — the overlap ($C \cap G$); OR — both circles ($C \cup G$).

Two important special cases

Mutually exclusive: A and B cannot both be true. $A \cap B = \emptyset$ (empty set). Example: "raining" and "not raining".

Exhaustive: A and A^c together cover everything. $A \cup A^c = \Omega$ (the whole outcome space). One of them must be true.

The events that make up a probability are necessarily mutually exclusive and collectively exhaustive.

Example: A and its complement A^c are always both mutually exclusive and exhaustive. So $P(A) + P(A^c) = 1$ — if you know the probability of one, you know the other.

All states (possible outcomes) of a variable must be mutually exclusive & exhaustive. Every variable in a probability tree must be set up so its branches cover all possibilities without overlap.

§4. Probability as a Measure — The Three Rules

What is a probability measure?

We've been using the word "probability" informally. Now we pin it down.

probability measure is a function that assigns a number to every event in the outcome space. Think of it as giving each region of the Venn diagram a **weight** (or "area"). The rules it must follow are simple:

Probability measure — a "measure" applied to events. Assigns a weight $P(A)$ to every event A , satisfying three axioms. **Everything else in probability — the probability algebra — follows from these three rules alone.**

The Three Axioms of Probability Algebra

Axiom 1: $P(\emptyset) = 0$ — The **impossible event** has probability zero. "Something that can never happen has no weight."

Axiom 2: $P(\Omega) = 1$ i.e. $P(\text{outcome space}) = 1$ — **Something must happen**. The total weight of the certain event = 1. Your certainty has to go somewhere. The outcome space itself is called **the certain event**.

Axiom 3: $P(A \cup B) = P(A) + P(B)$ if $A \cap B = \emptyset$ — **Non-overlapping events add**. If A and B cannot both happen, the probability of one or the other is just the sum. This is called **finite additivity**.

The Three Axioms — Rules Every Probability Must Obey

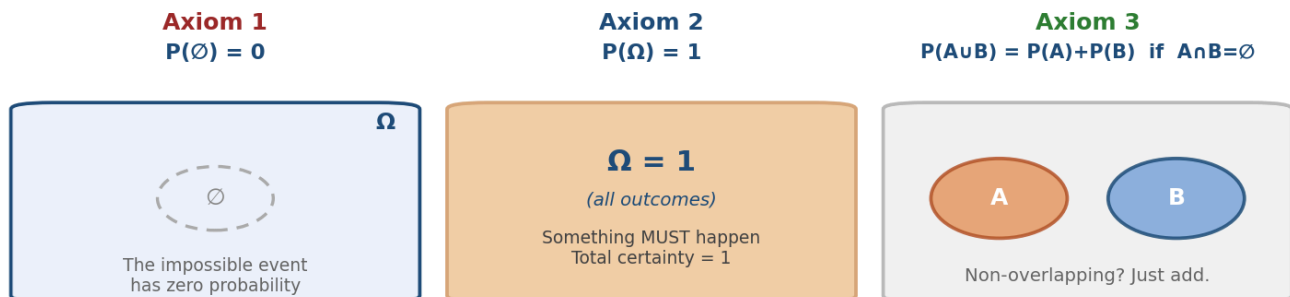


Figure 5. The three axioms visualized. Axiom 1: $\emptyset$ has no weight. Axiom 2: Ω has total weight 1. Axiom 3: non-overlapping events simply add.

Why only three? These three rules are the *minimum* needed to make probability coherent. Every formula you'll use — the complement rule, the Law of Total Probability, Bayes' theorem, the Chain rule — is a **theorem** derived from these three alone. You don't need to memorize more rules if you understand these.

What follows immediately from the axioms

Complement rule: Since A and A^c are mutually exclusive (Axiom 3) and together fill Ω (Axiom 2):

$$P(A^c) = 1 - P(A)$$

Example: $P(\text{not coffee drinker}) = 1 - P(\text{coffee drinker}) = 1 - 0.20 = 0.80$

General OR rule: When A and B do overlap:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Example: $P(C \cup G) = 0.20 + 0.60 - 0.04 = 0.76$

§5. Computing Probabilities — Conditioning on Events

Conditional Probability — the counting definition

When all point events are equally likely (fair coin, fair die, well-shuffled deck), computing probability is just counting:

Counting definition: $P(A|B) = \#(\text{point events in } A \cap B) / \#(\text{point events in } B)$ and $P(A \cap B) = P(A|B) \cdot P(B)$

Example: Roll a fair die. $\Omega = \{1, 2, 3, 4, 5, 6\}$, all equally likely. $P(\text{even}) = \#\{2, 4, 6\} / \#\{1, 2, 3, 4, 5, 6\} = 3/6 = 1/2$
 $P(\text{even} | \text{greater than 3}) = \#\{4, 6\} / \#\{4, 5, 6\} = 2/3$

Conditional probability — zooming in

The counting formula works when all outcomes are equally likely. All probabilities, in concept, are conditioned on ε — $P(A)$ means $P(A | \varepsilon)$. It is assumed that ε is always present, though for convenience we sometimes leave it out. Note that in our Coffee/College example the four outcomes have different probabilities (0.04, 0.16, 0.56, 0.24), because they are made of compound events — each one is a conjunction of two distinctions. We need a definition that works always:

Conditional probability $P(A|B)$ — read: "the probability of A **given** that B has occurred" — is the most important single idea in this course. Here is the intuition:

Before you know anything: $P(A)$ is your probability for A given only your background knowledge.

After learning B happened: $P(A|B)$ is your **updated** probability for A. Learning B rules out everything outside B, and you re-distribute your certainty only among the point events inside B.

Definition: $P(A|B) = P(A \cap B) / P(B)$ (whenever $P(B) > 0$)

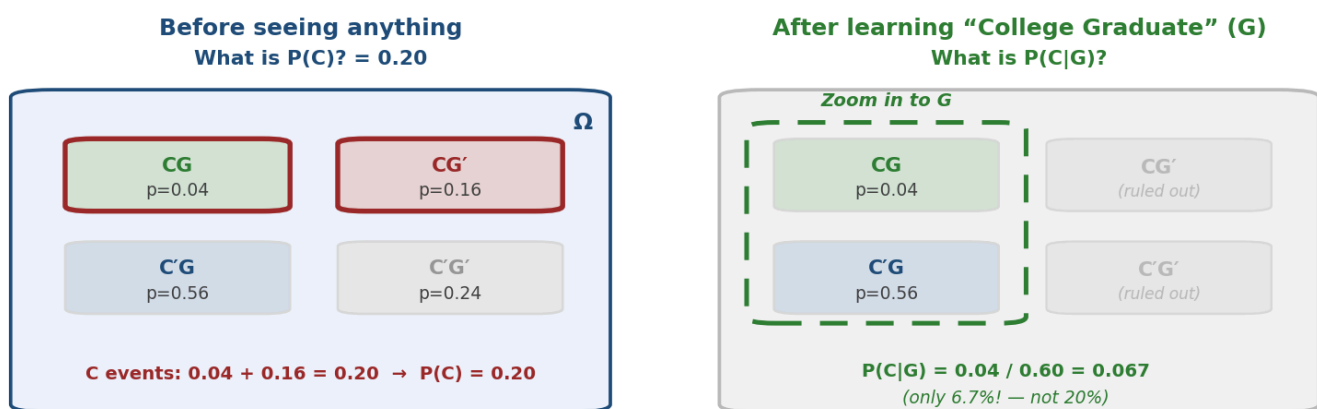


Figure 6. Left: before conditioning — $P(C) = 0.20$, using all 4 cells. Right: after learning "College Graduate" — we zoom into the G column only, and $P(C|G) = 0.04/0.60 = 0.067$. Learning G dramatically changed our belief about C.

Note — all probabilities are conditional: When we write $P(A)$, we really mean $P(A | \varepsilon)$ — the probability of A given your current background knowledge. Probability never exists in a vacuum.

Coffee / College example: $P(C|G) = P(CG) / P(G) = 0.04 / 0.60 = \mathbf{0.067}$ Knowing someone is a college graduate drops your probability they drink coffee from 20% to 6.7%. The two events are **not independent**.

The chain rule — how leaf probabilities are computed

Rearrange the conditional probability definition and you get the chain rule:

Chain rule: $P(A \cap B) = P(A|B) \cdot P(B)$ This is how you compute **each leaf probability in a tree**: multiply the branch probabilities along the path.

Example: $P(CG) = P(G|C) \cdot P(C) = 0.2 \times 0.2 = \mathbf{0.04}$ $P(C'G) = P(G|C') \cdot P(C') = 0.7 \times 0.8 = \mathbf{0.56}$

The Law of Total Probability

Sometimes you want $P(B)$ but you only know the conditional probabilities $P(B|A_i)$. The Law of Total Probability tells you how to combine them:

Law of Total Probability: $P(B) = P(B|A_1)P(A_1) + P(B|A_2)P(A_2) + \dots$ where the A_i form a **partition** of Ω : mutually exclusive ($A_i \cap A_j = \emptyset$) and exhaustive ($\cup A_i = \Omega$).

In a probability tree, this is simply: **sum all the leaf probabilities that lead to event B**. The A_i are the first-level branches — the partition of Ω is provided by the A_i .

Example: $P(G) = P(G|C) \cdot P(C) + P(G|C') \cdot P(C') = 0.2 \times 0.2 + 0.7 \times 0.8 = 0.04 + 0.56 = 0.60$

§6. Independence — Does One Thing Tell You About Another?

The intuition

Suppose you flip a coin and also check whether it will rain today. Does the coin result tell you anything about rain? Of course not — they have nothing to do with each other. We say they are

independent.

Now suppose you learn that the next person who walks in is a college graduate. Does that change your probability they drink coffee? From the numbers in our example: yes, it drops from 20% to 6.7%. Coffee drinking and college graduation are

dependent.

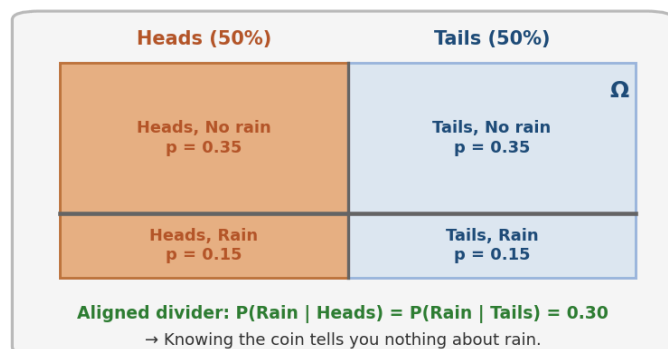
The formal definition

Independence $A \perp\!\!\!\perp B$: $P(A|B) = P(A) \iff P(A \cap B) = P(A) \cdot P(B)$ or equivalently: $P(A) \cdot P(B) = P(A \cap B)$ Events A and B are independent if knowing B gives you **no new information** about A — your probability for A is exactly the same whether or not you know B occurred.

Independence: does knowing one thing change your probability for another?

INDEPENDENT ($A \perp\!\!\!\perp B$)

Aligned partitions $\rightarrow$ equal conditional probabilities



DEPENDENT (knowing one DOES matter)

Slanted partitions show unequal conditional probabilities

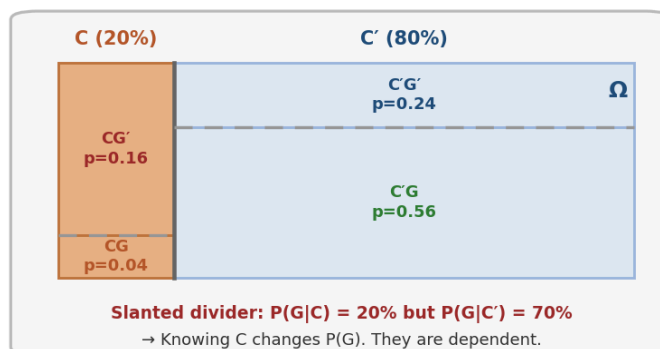


Figure 8. The outcome space drawn as a partitioned Ω . Left: independent events (coin flip and rain) — the Rain divider sits at the same height whichever way the coin lands (aligned partitions), so $P(\text{Rain}|\text{Heads}) = P(\text{Rain}|\text{Tails})$ and knowing one tells you nothing about the other. Right: dependent events (Coffee C and College Graduate G) — the divider is slanted (different heights under C and C'), so $P(C|G) = 0.067 \neq P(C) = 0.20$.

Two equivalent forms

Form 1: $P(A|B) = P(A)$ — the conditional probability equals the unconditional probability. Learning B doesn't move your needle on A.

Form 2: $P(A \cap B) = P(A) \cdot P(B)$ — the joint probability factors into a product. This is often easier to check numerically.

Important: Independence $\neq$ mutual exclusivity. If A and B are mutually exclusive and both have positive probability, they are actually *dependent*: learning A tells you immediately that B did *not* happen. Mutual exclusivity is a statement about whether events can *co-occur*; independence is a statement about whether they carry *information* about each other.

Symmetry

Independence is **symmetric**: if A is independent of B, then B is independent of A. Knowing A gives you no information about B, and knowing B gives you no information about A. You cannot have one-way independence. Similarly with dependence: If A is dependent on B then B is dependent on A.

Probability Algebra: Rules That Follow From the Three Axioms

Rule	Formula	Example with Coffee/College
Complement	$P(A^c) = 1 - P(A)$	e.g. $P(\text{not coffee}) = 1 - 0.20 = 0.80$
Law of Total P	$P(B) = \sum_i P(B A_i) \cdot P(A_i)$	e.g. $P(G) = P(G C) \cdot P(C) + P(G C') \cdot P(C') = 0.60$
Conditional P	$P(A B) = P(A \cap B) / P(B)$	e.g. $P(C G) = 0.04 / 0.60 = 0.067$
Chain Rule	$P(A \cap B) = P(A B) \cdot P(B)$	e.g. $P(CG) = 0.2 \times 0.2 = 0.04$
Independence	$P(A B) = P(A) \iff A \text{ indep } B$	e.g. Coin flip indep Today's weather

Figure 7. Summary of all probability algebra rules, each with a concrete Coffee/College example.

Quick Reference — All the Key Terms

event — a statement / proposition that includes a distinction at a point & time (true or false when that moment arrives), comprising two possible outcomes

point event = elemental outcome — the most specific possible event; a leaf in the tree

outcome space = sample space Ω — the complete set of all possible point events

event algebra — OR ($\cup$), AND ($\cap$), NOT (complement) operations on events

probability measure — a function that assigns weights to events, satisfying the three axioms

Axiom 1: $P(\emptyset) = 0$ | **Axiom 2:** $P(\Omega) = 1$ | **Axiom 3:** $P(A \cup B) = P(A) + P(B)$ if $A \cap B = \emptyset$

complement: $P(A^c) = 1 - P(A)$ (also written $P(A') = 1 - P(A)$)

counting definition: $P(A|B) = \#(A \cap B) / \#(B)$ — when all outcomes are equally likely

conditional probability: $P(A|B) = P(A \cap B) / P(B)$ — probability of A given B is known

chain rule: $P(A \cap B) = P(A|B) \cdot P(B)$ — multiply branch probabilities along a path in the tree

Law of Total Probability: $P(B) = \sum_i P(B|A_i) \cdot P(A_i)$ — sum over a partition of Ω provided by A

independence ($A \perp\!\!\!\perp B$): $P(A|B) = P(A)$, or equivalently $P(A \cap B) = P(A) \cdot P(B)$

distinction — a description, such as a property, that partitions all possible point events in the outcome space into 'crisp' subsets that are mutually exclusive and exhaustive

variable — the symbol, e.g. a letter, used to represent an event, or a measurable quantity

probability tree — a tree with events as nodes and leaves as outcomes, showing the dependence of outcomes on events

Based on Alvin W. Drake, Fundamentals of Applied Probability Theory, §1.2--1.3 | J. Blitzstein & J. Hwang, Introduction to Probability (Chapman & Hall / CRC, 2nd ed., 2019)